which the light enters the inner chamber. Iris is made of two types of muscles the **dilator papillae** (the radial muscle) and the **sphincter papillae** (the circular muscle).In the bright light, the circular muscle in the iris contract; so that the size of pupil decreases and less light enters the eye. In dim light, the radial muscle in the iris contract; so that the pupil size increases and more light enters the eye. Smooth muscle present in the ciliary body is called the **ciliary muscle** which alters the convexity of the lens for near and far vision. The ability of the eyes to focus objects at varying distances is called **accommodation** which is achieved by **suspensory ligament, ciliary muscle** and **ciliary body**. The

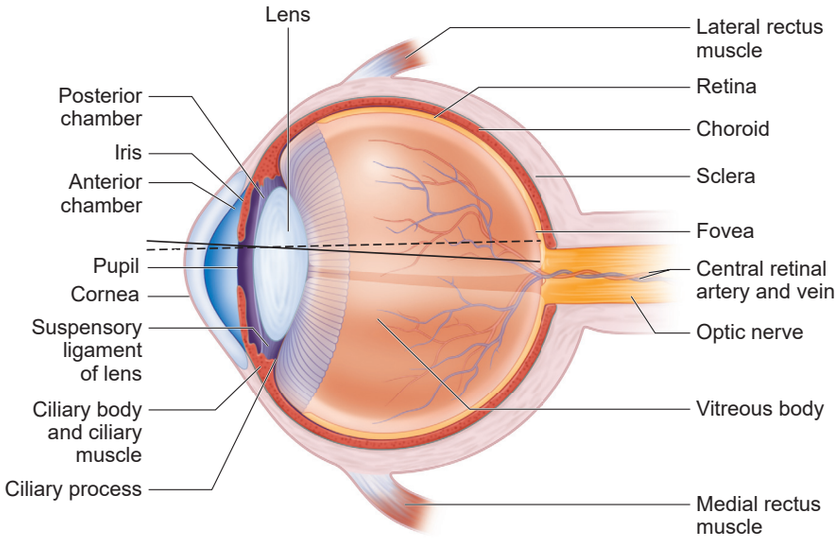


Figure 10.14 L.S. of the eye



Figure 10.15 Rod and Cone cells

suspensory ligament extends from the ciliary body and helps to hold the lens in its upright position. The ciliary body is provided with blood capillaries that secrete a watery fluid called **aqueous humor** that fills the anterior chamber.

Retina forms the inner most layer of the eye and it contains two regions: A sheet of **pigmented epithelium** (non visual part) and **neural visual regions.** The neural retina layer contains three types of cells: photoreceptor cells – **cones and rods** (Figure 10.15 and Table 10.4), **bipolar cells and ganglion cells.** The yellow flat spot at the centre of the posterior region of the retina is called **macula lutea** which is responsible for sharp detailed vision. A small depression present in the centre of the yellow spot is called **fovea centralis** which contains only cones. The optic nerves and the retinal blood vessels enter the eye slightly below the posterior

Table 10.4 Differences between rod and cone cells

Rod cells	Cone cells
Rods are responsible for vision in dim light	The cones are responsible for colour vision and works best in the bright light.
The pigment present in the rods is rhodopsin, formed of a protein scotopsin and retinal (an aldehyde of vitamin A)	The pigment present in the cones is photopsin, formed of opsin protein and retinal.
There are about 120 millions rod cells	There may be 6-7 millions cone cells
Rods are predominant in the extra fovea region	Cones are concentrated in the fovea region

XI Std Zoology Chapter-10 EM.indd 191 30/12/2022 13:55:38